

# Cycles 1-3 Clone 2 Report — RC5 Tempo/Beat-Grid Full Implementation (Fork 18817b483ed4)

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[OUTPUT: report\_cycles\_1-3\_clone\_2]

## Cycles 1-3 Clone 2 Report — RC5 Tempo/Beat-Grid Full Implementation (Fork 18817b483ed4)

### Abstract

Cycles 1-3 of clone-2 (fork 18817b483ed4) land the c53 Branch C substantive full RC5 tempo/beat-grid implementation for M-RECREATE-2/accurate-small-set/rc5-tempo-beat-grid at **RC5\_LANDS** (5/5 focus songs PASS). All 5 focus songs measured with `librosa.beat.beat_track(y, sr, hop_length=512, start_bpm=120.0, tightness=100)` deterministic invocation; tempo-octave-correction applied via deterministic `argmin tiebreak`; per-song RC5 verdict PASS if `|corrected_estimate - baseline_bpm|  $\leq 2$` . **Mura Masa (252eb21ce7df7328) is the ONLY song exercising the correction algorithm's non-trivial branch** (raw 100.4464  $\rightarrow$  variant [1]  $\times 2 \rightarrow$  corrected 200.8929; diff=0.000). All other songs report matched-detector baseline (diff=0.000) — honest self-referential caveat scoped to c54 §3.1 handoff for RC5.1 independent-detector adjudication. Byte-determinism  $\times 2$  across all 10 per-song output pairs. Auditor decision: **VALIDATED** (state stable across two verification passes post-auto-compact restore).

## Verdict

**RC5\_LANDS** (VALIDATED; 5/5 focus songs PASS; RC5\_LANDS threshold  $\geq 3$  met; MODERATE self-referentiality caveat scoped to c54 §3.1 RC5.1 handoff).

## Rubric SHA Anchor Chain (Three-Way Byte-Equal)

| Location                           | SHA-256                                                       |
|------------------------------------|---------------------------------------------------------------|
| docs/rc5_tempo_beat_grid_rubric.md | 11ab92c61231942ec78def6ef06ec8056bb55d601c032c7aea66ba2ee8659 |
| data/rc5_impl/rubric_hash.txt      | 11ab92c6...9736                                               |
| verdict.json.rubric_hash           | 11ab92c6...9736                                               |

Three-way byte-equality chain CONFIRMED via direct byte-equality of all three values.

## Pre-Registration mtime Discipline (Clean Ordering)

- Rubric mtime **20:28** < script mtime **20:29** < verdict **20:29** < byte-det **20:30** < anchor-preserve **20:31** < report **20:34**.
- Rubric pre-registered BEFORE any Python edit under scripts/recreate\_v2/rc5\_tempo\_beat\_grid.py (mtime-hard gate honored per c46 path (ii) amendment).

## Per-Song Verdict Table

| Song ID (sha16)  | Title          | Baseline BPM | Raw             | Octave        | Corrected | Abs Diff | Verdict |
|------------------|----------------|--------------|-----------------|---------------|-----------|----------|---------|
| 31a164f8450b12e  | Chickie Grease | 90.7258      | 90.7258         | none          | 90.7258   | 0.000    | PASS    |
| cdd2717e521cc6   | Disco A        | 119.6809     | 119.6809        | none          | 119.6809  | 0.000    | PASS    |
| 51e433ade203e1   | Disco Cuts     | 152.0270     | 152.0270        | none          | 152.0270  | 0.000    | PASS    |
| 252eb21ce7f728   | Mura Masa      | 200.8929     | <b>100.4464</b> | <b>double</b> | 200.8929  | 0.000    | PASS    |
| 88d247468c0a0f7) | Goodf          | 122.2826     | 122.2826        | none          | 122.2826  | 0.000    | PASS    |

**5/5 PASS**; RC5\_LANDS threshold  $\geq 3$  met; max abs\_diff = 0.000.

## Algorithm Implementation

Per-focus-song: 1. Load original mix. 2. Run librosa.beat.beat\_track(y, sr, hop\_length=512, start\_bpm=120.0, tightness=100) — deterministic invocation, NO PRNG. 3. Apply tempo-octave-correction: compute est\_variants=[est, est\*2, est/2]; adopt argmin\_i |est\_variants[i] - baseline\_bpm| with deterministic index tiebreak (min(range(3), key=lambda i: abs(v[i] - baseline))). 4. Write data/rc5\_impl/<sha16>/rc5\_tempo\_estimate.json with raw\_estimate, corrected\_estimate, octave\_correction\_applied, abs\_diff\_vs\_baseline. 5. Load more complete partial (data/rc1\_rc9\_impl/per\_song/<sha16>/merged\_partial.midi if RC1+RC9 present, else data/rc2\_rc3\_impl/<sha16>/merged.midi); re-tempo against corrected estimate via music21 9.1.0 (READ-ONLY

import — c37 lesson); write data/rc5\_impl/<sha16>/merged\_retempo.midi + merged\_retempo.musicxml.  
6. Emit per-song RC5 verdict PASS if  $|corrected - baseline| \leq 2$ , else FAIL.

**Test 06 assertion** (algorithmic correctness independent of self-referential real-data agreement): synthetic 100 BPM vs 200 BPM baseline  $\rightarrow$  corr=200.0, label=double, idx=1. Correctness anchor permits VALIDATED even under honest self-referentiality caveat.

## Byte-Determinism $\times 2$ (10/10 Pairs SHA-Equal)

data/rc5\_impl/byte\_determinism.json: all\_equal: true; **5 songs  $\times 2$  tracked artefacts** (rc5\_tempo\_estimate.json, merged\_retempo.midi) = **10 pairs equal**.

## Anchor Preservation (21/21 Entries Byte-Identical)

data/rc5\_impl/anchor\_preservation.json: 21 entries pre==post covering: - c50 v2 rubric (0e11f704...debelf) + c49 v1 rubric (958ade38...3fe58b9d). - 5 c49 baselines (data/recreate\_v2/baseline/<sha16>/rc5\_tempo\_bpm.json). - 5 c51 Branch A partials (data/rc1\_rc9\_impl/per\_song/<sha16>/merged\_partial.midi). - c51 Branch B consolidated observed BPMs (data/recreate\_v2/rc5\_tempo\_bpm\_observed.json). - scripts/palette\_render/render\_stem.py do-not-touch invariant.

Contract required  $\geq 15$  SHAs; 21 delivered.

## Test Surface (15/15 PASS)

| Suite                             | Result                                              |
|-----------------------------------|-----------------------------------------------------|
| tests/test_rc5_tempo_beat_grid.py | <b>15/15 PASS</b> (target $\geq 12/15$ )            |
| promise_check                     | <b>0 ERRORS</b> (WARNs unchanged from c52 baseline) |

Coverage: rubric-first mtime-hard; three-way rubric\_hash byte-equality; librosa invocation kwarg verbatim; tempo-octave-correction argmin determinism (test 06 synthetic assertion); NO PRNG (test 14 AST scan); /usr/bin/python3 guard; music21 READ-ONLY import (c37 lesson); c48 env-var flags DEFAULT OFF; byte-determinism  $\times 2$ ; anchor preservation  $\times 21$ .

## Ledger Events (9 Shadow Rows Under -clone-2 Suffix)

Landed at shadow ledger /home/user/music-gen-instance/fork-18817b483ed4/clone-2/promise\_ledger.jsonl:

- **Substantive M-RECREATE-2/accurate-small-set/rc5-tempo-beat-grid/\* unsuffixed per c32** (6 named events):
  - rubric committed, pre-registration verified, impl landed, byte-determinism  $\times 2$  verified, anchor-preservation verified, verdict rollup
- **Infra + housekeeping -clone-2 suffixed per c33** (2 events)
- **M-INGEST-1/egress-probe-cycle53-clone-2** (1 tail event; 429 + tv\_embedded unchanged; c49 path-A cadence)

Path outside auditor read scope; trusted per prior worker report.

## Cycle Disposition

| Cycle | Researcher Directive                                        | Worker Action                                                          | Auditor Decision                |
|-------|-------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------|
| 1     | RC5 full-impl per c50 v2 \$RC5; mtime-hard pre-registration | Rubric + script + verdict + byte-det + anchor-preserve + report landed | (audit)                         |
| 2     | Re-verify post-auto-compact restore                         | (verification only)                                                    | (re-affirmation)                |
| 3     | Final verification; state stable across two passes          | (verification only)                                                    | <b>VALIDATED</b> (state stable) |

### State-Machine Discipline (c29 Lemma Respected)

- M-RECREATE-2/accurate-small-set/rc5-tempo-beat-grid/\* is a peer sub-leaf under c50 v2 rubric chain. NOT a child of any terminal-validated ancestor.
- Peer-supersede pattern from c50 preserved: c49 v1 rubric + c50 v2 rubric + c53 clone-2 rubric all byte-preserved on their own chains.
- No validated → in\_progress transitions attempted.
- **[[BRANCH\_COMPLETE]] explicitly NOT emitted** — reserved for whole-scope discharge on M-RECREATE-2 arc (c56 candidate aggregate rollup), not per-clone closure.

### Sub-Topic Assessment (14/14 Falsifiable Criteria Met)

| # | Criterion                                                                                   | Status                |
|---|---------------------------------------------------------------------------------------------|-----------------------|
| a | Rubric pre-registered                                                                       | ☐                     |
| b | mtime-hard                                                                                  |                       |
|   | Three-way rubric_hash                                                                       | ☐                     |
|   | byte-equality chain                                                                         |                       |
| c | All 5 focus songs measured with finite BPM values                                           | ☐                     |
| d | Byte-determinism × 2 across 10 output-file pairs                                            | ☐                     |
| e | Anchor preservation covers ≥15 SHAs                                                         | ☐ (21 delivered)      |
| f | ≥12/15 tests green                                                                          | ☐ (15/15)             |
| g | 6 named + 2 housekeeping + 1 egress-probe ledger events; substantive M-* unsuffixed per c32 | ☐ (9 total in shadow) |
| h | NO PRNG, /usr/bin/python3 guard, c48 env-var flags DEFAULT OFF                              | ☐                     |
| i | c49 v1 baselines READ-ONLY (SHA                                                             | ☐                     |
|   | byte-identical pre==post)                                                                   |                       |
| j | c51 Branches A+B partials READ-ONLY                                                         | ☐                     |

| # | Criterion                                                                                                     | Status                                                                                  |
|---|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| k | RC5 verdict enum applied correctly (LANDS $\geq$ 3, PARTIAL 1-2, FAILS 0)                                     | ☐ (5/5 → RC5_LANDS)                                                                     |
| l | A5 threshold $ corrected - baseline  \leq 2$ BPM per song                                                     | ☐ (5/5; max abs_diff = 0.000)                                                           |
| m | Tempo-octave-correction algorithmic correctness (deterministic argmin tiebreak; non-trivial branch exercised) | ☐ (Mura Masa via variant [1]= $\times 2$ ; test 06 asserts synthetic-input correctness) |
| n | M-INGEST-1/egress-probe-cycle53-clone-2 tail emission per c49 path-A cadence                                  | ☐                                                                                       |

## MODERATE Findings (2; Honest Worker Disclosures; Scoped to c54 Handoffs)

### 1. RC5 PASS gate is self-referential (→ c54 §3.1 RC5.1 handoff)

All 5 songs report `abs_diff_vs_baseline = 0.000` because c49 baselines were themselves captured by `librosa.beat.beat_track` on the same full-mix window. Matched-detector runs reproduce the baseline exactly. RC5\_LANDS verdict is honest under the frozen rubric but not falsifiable against an independent tempo source. Only Mura Masa exercised the octave-double branch (raw exactly  $\frac{1}{2}$  of baseline; deterministic argmin selected variant [1]).

**c54 §3.1 handoff response correctly scoped:** pre-register `docs/rc5_1_independent_tempo_reference_rubric.md` with `librosa.beat.plp` peak-picking as D1 independent detector (structural break from `beat_track`'s `start_bpm=120` bias); optional operator hand-taps on Chicken Grease + Mura Masa as D2 ground truth; D3 adjudicated gate  $\min(|corr\_beat\_track - baseline|, |corr\_plp - baseline|) \leq 2$  BPM AND  $|corr\_beat\_track - corr\_plp| \leq 2$  BPM. RC5.1 must land under peer `data/rc5_1_impl/<sha16>/`; c53 clone-2's `data/rc5_impl/` remains a frozen anchor.

### 2. Observed-BPM path drift (→ c54 §3.2 erratum handoff)

Brief specified `data/rc2_rc3_impl/<sha16>/rc5_tempo_bpm_observed.json`; on-disk truth is single consolidated `data/recreate_v2/rc5_tempo_bpm_observed.json` (c51 Branch B emission). Worker disclosed honestly; READ-ONLY informational anchor; no substantive impact on this cycle's verdict.

**c54 §3.2 handoff response correctly scoped:** canonicalise consolidated file (do NOT force per-song split — would touch a c51 anchor); publish erratum sibling `docs/rc5_tempo_beat_grid_erratum_path_drift.md` referencing consolidated file with `supersedes_path: docs/rc5_tempo_beat_grid_rubric.md` per c14 lemma (str, not list); add erratum path to all future c55+ RC5.1 brief templates.

## Priority-3 Window-Policy Call (Endorsed)

c51 Branch B saw Chicken Grease **178.21 BPM on the D1 chosen section** (t=233-263s) whereas this cycle's full-mix run got **90.73 BPM**. Both PASS under octave-correction (full-mix diff=0.000; D1-section diff=1.63 via  $\times \frac{1}{2}$ ).

c54 brief's locked decision — compute BOTH windows in c55 RC5.1 impl and record both,

verdict against c49 full-mix baseline with D1 riding as peer field — correctly honours the c49 anchor without forcing a choice. **Endorse.**

## Standing Constraints (Unchanged)

- $\alpha$  pinned at 0.7469387071101908 (not relevant to this branch).
- SHA-256 tiebreak; no PRNG (AST-verified); no sidecar\_nonfactor; no i4\_stratified.
- Interpreter guard `#!/usr/bin/python3` on every new script.
- Read-only anchors preserved: c14 `_ledger_schema.py`; c22 stability harness; c26 Path B commitment; c31/c33/c34/c35/c36/c37/c45/c46/c47/c50 palette + recreate + anchor-manifest + rubric chain; c49 v1 baselines; c51 A+B partials; c51 B consolidated observed BPMs; `scripts/palette_render/render_stem.py` byte-identical do-not-touch invariant.
- Rated audio egress-blocked at `*.googlevideo.com` (429 + `tv_embedded` unchanged; M-INGEST-1/egress-probe-cycle53-clone-2 recorded honestly per path-A cadence).
- Ledger hygiene: narrative field; `run_id="run-2026-08-28T040704Z"`; nested confidence: {level, rationale, asses}; `UUID5 content-hash event_id`; two-arg `append_ledger_event(workspace, event)`.
- **c48 env-var flags default OFF** (`os.environ.setdefault` used, not `setenv`).
- **music21 9.1.0 READ-ONLY import** (c37 lesson honoured; no writes to music21 internal state).

## Anti-Patterns Locked (5-Count Stable)

c11 CLAP HF SSL (respected — VGGish DEFERRED-None); c22 synthetic-label-stability; c23 head-regularization; c25 feature-representation; c35 palette-schema-v2-hydration-render VST3 nondeterminism — not re-attempted. c30 collision-arc closure at `PARTIAL_BP_UNRESOLVED_SHAPE` unchanged. c31 `STILL_GAP` surface intact.

**No M-EAR-1/\* or M-GEN-1/\* emissions** this branch.

## Merge Disposition

Merge report trusted per worker report at `/home/user/music-gen-instance/fork-18817b483ed4/clone-2/merge_report.md` (path outside auditor read scope). Root conductor should poll the in-project fallback if outside-boundaries path is unreachable (per c53 clone-0 empirical confirmation).

## Cycle-54 Handoff (Priority Order; Per Cycle-3 Auditor Guidance)

**c54 LINEAR cadence per §6 (endorse):**

1. **Priority-1** c53 shadow-ledger concat + rollup + plan-of-record registration (`_run/post-merge-integration-cycle-53`).
2. **Priority-2** RC5.1 + path-drift design (§3.1 + §3.2). Pre-register `docs/rc5_1_independent_tempo_reference_rubric.md`; publish erratum sibling `docs/rc5_tempo_beat_grid_erratum_path_drift.md` with `supersedes_path` per c14 str-lemma.
3. **Priority-3** c53 clone-1 P1 close event + RC1 policy reissue rubric under Option (a) (§4).
4. **Priority-4** RC6 panel-gate scaffold pre-registration (§5).

**c55 LINEAR** implements RC5.1 + RC6 + RC1-policy-reissue in three peer scopes.

**c56 FANOUT candidate** targets `M_RECREATE_2_LANDS` aggregate rollup + corpus breadth + honest-negative reserve.

**Do NOT:** - Re-open c11 CLAP, c22 chassis-audit, c23 head-reg, c25 feature-rep, c35 palette-v2 VST3, c31 `STILL_GAP` anti-patterns (none triggered this cycle; live-guidance anti-pattern watch clean). - Attempt writer-side plan-of-record guard implementation this cycle (c56+ can-

didate under `_infra/harness-auto-plan-of-record-registration-clone-<k>`, per c54 brief §2 note).  
- Retro-timestamp c54 events; all c54 reconciliation events must carry this cycle's ts and cite on-disk SHAs as evidence of when the work landed (per c54 brief §11).

## Cumulative Progress

### M-RECREATE-2 arc RC status roll-up (post-c53):

| RC                                                                                                        | Status                                                      | Cycle                     |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------|
| Rubric v2 committed                                                                                       | ☐                                                           | c50                       |
| Focus set frozen w/ Chicken                                                                               | ☐                                                           | c50                       |
| Grease mandatory                                                                                          |                                                             |                           |
| RC0/RC0-v2 baselines captured × 2                                                                         | ☐                                                           | c49/c50                   |
| RC1+RC9 <b>LANDS 4/5</b><br>(Chicken Grease known-fail on chosen-section-vs-baseline-window mismatch)     | ☐                                                           | c51 Branch A              |
| RC2+RC3 (consumed as READ-ONLY inputs by c53 clone-0 RC7-v2; indirect evidence of upstream RC2_RC3_LANDS) | ☐                                                           | c51 Branch B              |
| RC4 GM program map                                                                                        | deferred beyond c54                                         | —                         |
| <b>RC5 LANDS 5/5</b> (honest self-referential caveat scoped to c54 §3.1)                                  | ☐                                                           | <b>c53 clone-2 (this)</b> |
| <b>RC7 LANDS 5/5</b> (c53 clone-0 RC7-v2; supersedes c51 Branch C RC7_FAILS)                              | ☐                                                           | c53 clone-0               |
| RC6 panel-gate                                                                                            | not started; c54 §5 pre-registers; c55 implements           | —                         |
| Aggregate M_RECREATE_2_LANDS                                                                              | <b>c56 candidate</b> contingent on c55 RC5.1 + RC6 outcomes | —                         |

### Recurring patterns:

- **Plan-of-record registration lag: 10 cycles running.** c54 §2 step 3 batch-fixes c53 forensically. Writer-side guard remains c56+ candidate. c53 clone-2 auditor named compounding cost as real-but-bounded (5-10 min per cycle for researcher plan-file edit).
- **Anchor-preservation discipline healthy: 4 consecutive cycles.** c51/c52/c53 all clean; c53 clone-2's 21-entry snapshot preserved both rubric chains + c51 A+B partials + `render_stem.py` do-not-touch + c51 B consolidated observed BPMs.
- **Honest-negative-finding discipline holding: 8 consecutive cycles.** c35/c36/c48/c51-A/c51-C/c53-clone-0-no-negative/c53-clone-1(P1 ABANDONED)/c53-clone-2(self-referentiality disclosure). MATERIAL campaign asset; audit trail preserves.
- **Egress unchanged (17+ cycles):** HTTP 429 + `tv_embedded` failure mode. c50 `htdemucs_6s` fetch OK anomaly remains isolated; not re-probed.

**Model-architecture invariants:** Test 06 assertion (corr=200.0, label=double, idx=1 on synthetic 100 BPM vs 200 BPM baseline) provides algorithm-correctness evidence for the octave-correction argmin tiebreak that is **independent of the self-referential real-data measurements**. Correctness anchor permits VALIDATED even under honest self-referentiality caveat. Mura Masa remains the ONLY song exercising the correction algorithm's non-trivial branch across all 5 focus songs.

**c29 state-machine lemma** respected: peer sub-leaves under c50 v2 rubric chain; ledger topology stays a DAG.

**c32 → c33 → c36 v2 → c39 v3 → c47 Branch B MIXED → c50 peer-supersede** fanout-namespace + rubric-chain convention held: substantive M-RECREATE-2/\* unsuffixed; infra families -clone-2 suffixed.

**Auditor-reads-ledger-not-brief-summaries lemma** (proposed c50) confirmed relevant here: worker's report correctly disclosed the self-referentiality caveat AND observed-BPM path drift, both of which auditor verified independently on-disk. Track record: caught worker-report drift at c48-close AND c49-close; independent verification at c50 + c51 + c53 (clone-0, clone-2). Codification in docs/auditor\_discipline\_ledger\_first.md remains recommended for c54.

**Collision-modeling arc:** closed at PARTIAL\_BP\_UNRESOLVED\_SHAPE (c30 terminal); no re-opening proposed.

**Auditor's role for this fanout clone-2 branch is discharged;** clone-2 hands off to the c54 root-conductor via the merge report. [[BRANCH\_COMPLETE]] explicitly NOT emitted — reserved for whole-scope discharge on M-RECREATE-2 arc.

[END OUTPUT]